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Review

Chronic kidney disease-induced muscle atrophy: Molecular mechanisms and promising therapies

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ABSTRACT

Chronic kidney disease (CKD) is a high-risk chronic catabolic disease due to its high morbidity and mortality. CKD is accompanied by many complications, leading to a poor quality of life, and serious complications may even threaten the life of CKD patients. Muscle atrophy is a common complication of CKD. Muscle atrophy and sarcopenia in CKD patients have complex pathways that are related to multiple mechanisms and related factors. This review not only discusses the mechanisms by which inflammation, oxidative stress, mitochondrial dysfunction promote CKD-induced muscle atrophy but also explores other CKD-related complications, such as metabolic acidosis, vitamin D deficiency, anorexia, and excess angiotensin II, as well as other related factors that play a role in CKD muscle atrophy, such as insulin resistance, hormones, hemodialysis, uremic toxins, intestinal flora imbalance, and miRNA. We highlight potential treatments and drugs that can effectively treat CKD-induced muscle atrophy in terms of complication treatment, nutritional supplementation, physical exercise, and drug intervention, thereby helping to improve the prognosis and quality of life of CKD patients.

1. Introduction

Chronic kidney disease (CKD) is a devastating chronic catabolic disease, and it is not a single disease but a general term for multiple diseases that damage the kidney and decrease the glomerular filtration rate (GFR) [1]. CKD is irreversible and progressive, and it is characterized by the progressive loss of kidney mass and function [2,3]. At present, GFR and albumin-creatinine ratio are two markers used to identify CKD [4]. CKD can be divided into five stages according to the GFR, and CKD, at the most severe end stage, is defined as chronic renal failure [5]. If CKD patients fail to receive timely and effective treatment in the first to third stages, CKD will progress to end-stage chronic renal failure and eventually lead to uremia and the complete loss of renal function, thus endangering the life of CKD patients [6]. In addition, CKD is accompanied by many complications, including some manifestations and symptoms of systemic organs, such as anemia [7], hyperkalemia [8], vascular calcification [9], mineral bone disease [10], muscle atrophy [11], metabolic acidosis [12], and cardiovascular disease [13], leading to poor quality of life in CKD patients. Currently, CKD has become a serious public health problem all over the world due to its high morbidity and

Abbreviations: CKD, chronic kidney disease; UPS, ubiquitin-proteasome system; ALP, autophagy-lysosome pathway; IGF, insulin growth factor; TLRs, Toll-like receptors; TGF-β, transforming growth factor-β; IRS-1, insulin receptor substrate 1; TNF-α, tumor necrosis factor-α; CRP, C-reactive protein; IL, interleukin; ROS, reactive oxygen species; RNS, reactive nitrogen species; SOCS-3, cytokine signaling inhibitory protein-3; IRF3, interferon regulatory factor 3; VDR, vitamin D receptor; IS, indoxyl sulfate.

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mortality [14].

Skeletal muscle is the largest tissue in the human body, comprising approximately 40 % of the body's total mass [15,16]. Skeletal muscle is an important organ that maintains many important functions of the human body, such as movement, visceral protection, metabolism, and respiration [17–19], which is essential for maintaining bodily integrity and activities of daily living. Skeletal muscle stem cells, also known as satellite cells, are primary stem cells in muscle tissue [20]. Satellite cells are not only the key mediator of muscle plasticity [21] but are also responsible for skeletal muscle growth, maintenance, and repair [22]. Skeletal muscle homeostasis is maintained by a balance between protein anabolism and catabolism [23,24]. However, under some pathological conditions, such as starvation, denervation, cachexia, weightlessness, diabetes, and renal failure, this balance is disrupted and skeletal muscle atrophy consequently occurs [19,25–28]. Muscle atrophy is characterized by muscle mass loss, activation of proteolytic pathways, and inhibition of protein synthesis pathways [18]. Recent studies have suggested that muscle atrophy is closely related to the activation of protein degradation pathways, namely, the ubiquitin–proteasome system (UPS) and autophagy-lysosome pathway (ALP), as well as the inhibition of the insulin growth factor (IGF)/PI3K/AKT/PKB-mTOR protein synthesis pathway [17,29–35].

Muscle atrophy and decreased muscular endurance are two important features of CKD. Satellite cells are usually in a quiescent state under normal conditions, but they can be activated by injury, exercise, or changes in the level of growth factors (such as IGF-1), which causes them to actively proliferate and differentiate into myotubes, forming muscle cells to compensate for muscle fiber loss [36]. In healthy individuals, this compensation is usually sufficient. However, in patients with muscle atrophy, the self-renewal capacity of satellite cells is considerably reduced [20]. CKD has been reported to impair the activation and differentiation of satellite cells [37] as well as produce low levels of myogenin and MyoD [38], and it has been shown to reduce muscle regeneration and repair capacity, eventually leading to muscle atrophy [5]. Furthermore, sarcopenia is a progressive and systemic skeletal muscle disease with accelerated loss of muscle mass and function [39]. Moreover, CKD patients with decreased exercise tolerance are more prone to developing sarcopenia, which is a common condition in all stages of CKD [40]. Sarcopenia also increases the risk of morbidity, mortality, and end-stage renal disease in patients with CKD [40,41]. In CKD, the pathophysiology of sarcopenia is complex. Although it is partly driven by aging, sarcopenia in CKD patients is possibly exacerbated by the accumulation of uremic toxins, inflammation, insulin resistance, malnutrition, oxidative stress, ubiquitination, and lack of physical activity [40]. In addition, the related mechanisms that promote CKD-induced muscle atrophy include impaired function of satellite cells, anorexia, vitamin D deficiency, mitochondrial disorder, hormone abnormality, excessive autophagy, impaired microRNA response, increased angiotensin II, increased myostatin, increased caspase 3, abnormal neuropeptide signal, intestinal flora imbalance, and metabolic acidosis [11,42]. Some possible treatment methods have been proposed for these mechanisms. Continuous research on these mechanisms not only provides new feasible ideas for the treatment of CKD-induced muscle atrophy but may also identify new treatments for CKD.

2. Mechanisms OF CKD-INDUCED muscle atrophy

The molecular mechanism of CKD-induced muscle atrophy is complicated. The current mainstream view is that CKD-induced muscle atrophy is closely related to inflammation, oxidative stress, insulin resistance, vitamin D deficiency, excessive angiotensin II, intestinal flora imbalance, anorexia, hemodialysis, metabolic acidosis, and uremia toxins. These factors can activate downstream proteolysis and inhibit protein synthesis, thereby causing skeletal muscle atrophy.

2.1. Molecular mechanisms of muscle mass loss

The proteolytic pathways mainly include UPS, ALP, Myostatin and caspase-dependent pathways while IGF-1/Akt is the major protein synthesis pathway in skeletal muscle (Fig. 1).

2.1.1. Protein degradation pathway

Muscle atrophy in CKD patients is mainly related to muscle proteolysis that is accelerated by the activated UPS. The UPS, the major intracellular protein hydrolysis system, is responsible for the degradation of misfolded or damaged proteins in skeletal muscle [19,43]. In the system, Ubiquitin, a short protein comprising 76 amino acids, can activate proteolysis [44]. The ubiquitination of proteins proceeds through a cascade of reactions that are catalyzed enzymes including ubiquitin activating (E1), conjugating (E2) and ligating (E3) enzymes [45]. These tagged proteins are subsequently recognized by the 26S proteasome, which consists of a barrel-shaped 20S proteolytic core and two 19S regulatory caps. These 19S regulatory caps can recognize and bind to ubiquitinated proteins to undergo specific proteolysis [46]. In the UPS, MuRF1 and MAFbx are two muscle-specific E3 ubiquitin ligases that are crucial for regulating muscle atrophy [45]. In addition, FoxO1, FoxO3, and FoxO4, as members of the FoxO family, modulate a series of atrophy-related genes, including MAFbx and MuRF1, thereby regulating skeletal muscle atrophy [47]. Therefore, the UPS is one of the main mechanisms underlying muscle atrophy.

The ALP is a physiological or pathological mechanism that plays an important role in maintaining protein balance, clearing damaged organelles, maintaining skeletal muscle mass, and stabilizing homeostasis [48]. Studies have found that FoxO3 induces the expression of many autophagy-related genes and enhances autophagy in skeletal muscle cells. However, overactivated autophagy accelerates muscle atrophy, and impaired autophagy leads to muscle degeneration [49,50]. Under catabolic conditions, such as CKD, defective IGF-1/PI3K/AKT signaling activates the ALP by disinhibiting FoxO activity [51], thereby accelerating protein degradation. The autophagy-related protein, Beclin 1, is a major determinant of autophagy initiation. FoxO3 regulates BNIP3,

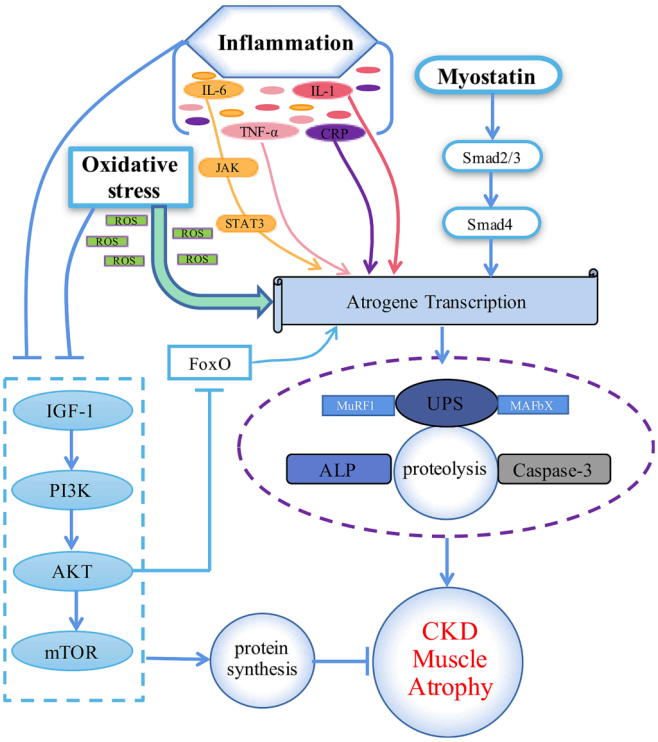

Fig. 1. Regulatory mechanisms of CKD-induced muscle atrophy.

which is required for autophagosome formation during muscle atrophy. Cathepsin L is a lysosomal protease responsible for degrading autophagosome contents. The expression of autophagy-related proteins, including Beclin 1, BNIP3, cathepsin L, and LC3, is increased in CKD patients, indicating an increase in ALP activity [11]. Overall, CKD induces skeletal muscle atrophy by upregulating the ALP.

Myostatin, an important member of the transforming growth factor-β (TGF-β) family, is a negative regulator of skeletal muscle growth, and it is secreted by skeletal muscle cells [52]. CKD-induced muscle atrophy is closely related to myostatin, which activates the UPS and inhibits satellite cell function [51,53]. In addition, myostatin binds to its activin type IIB receptor to activate the Smad2/Smad3 signaling pathway in muscle, phosphorylate AKT, reduce the phosphorylation of the FoxO transcription factor family, and increase the expression of MuRF1 and MAFbx, thereby resulting in enhanced muscle protein catabolism [47,54]. Inhibition of myostatin improves satellite cell function and inhibits protein degradation, thereby alleviating or inhibiting muscle atrophy [55]. Therefore, myostatin can be used as one of the methods to treat CKD-induced muscle atrophy.

Although the UPS can rapidly degrade actin or myosin, it cannot degrade large and complex structures, such as actomyosin and myofibrils [56]. At this point, the role of caspase 3 is of vital importance. Caspase 3 is activated under catabolic conditions (such as CKD), acting on the complex structure of muscle protein, cutting it into a substrate for the UPS [5], triggering protein degradation, and causing muscle atrophy. Caspase 3 stimulates UPS-mediated protein degradation in muscle by directly stimulating the proteolytic activity of proteasomes [55]. Therefore, caspase 3-mediated proteolysis is regarded as another important process of CKD-induced muscle atrophy, providing a potential target for the treatment of muscle atrophy.

2.1.2. Protein synthesis pathway

IGF-1 is a key growth factor that regulates skeletal muscle anabolic and catabolic pathways. IGF-1 promoting muscle protein synthesis via activating PI3K/AKT/mTOR pathway [57,58].The inactivation of IGF-1 receptors impairs muscle growth, triggering a reduction in the number and diameter of muscle fibers [59]. mTOR is usually assembled into complexes such as mTOR complex 1/2 (mTORC1/2). mTORC1 catalyzes the phosphorylation of multiple targets including its effector 70-kDa ribosomal protein S6 kinase (p70S6K) and eukaryotic translation initiation factor 4E binding protein 1. This leads to increased protein translation and synthesis, which subsequently promotes muscle growth [60,61].

2.2. Regulatory mechanisms of CKD induced muscle atrophy

Inflammation, oxidative stress and mitochondrial dysfunction are identified as key factors in CKD induced muscle atrophy (Fig. 1).

2.2.1. Inflammation

Recent research has well demonstrated that chronic inflammation affects skeletal muscle strength and mass [62,63]. Among many inflammatory cytokines, the main ones are tumor necrosis factor-α (TNF-α), C-reactive protein (CRP), interleukin (IL)-6, and IL-1β. TNF-α is a key stimulator of chronic inflammation that triggers skeletal muscle contractile dysfunction. TNF-α inhibits protein synthesis via the IGF1/AKT pathway, thereby triggering muscle atrophy. Sustained elevation of IL-6 aggravates muscle atrophy by reducing muscle anabolism and damaging energy homeostasis, and it also directly mediates muscle catabolism [23,25,64,65]. Increased IL-6 stimulates muscle protein degradation by interfering with insulin/PI3K/AKT signaling [27,55]. In addition, inflammation activates myostatin, thereby accelerating UPS-mediated catabolism and promoting muscle atrophy.

In CKD patients, inflammatory markers that are at high circulating levels include IL-6, TNF-α, and CRP [55,66,67]. The inflammatory state of CKD is associated with muscle atrophy because inflammatory cytokines play an important role in activating the nuclear factor-κB pathway, which induces MuRF1 expression, thereby activating the UPS [5,68]. Changes in IL-6 and CRP levels are related to SOCS-3, and increased SOCS-3 expression in muscle tissue causes the loss of muscle proteins [69,70]. High SOCS-3 levels reduce insulin receptor substrate 1 (IRS-1) levels and inhibit intracellular insulin signaling, leading to the activation of caspase 3 and UPS, which in turn promotes muscle atrophy [55,71]. Additionally, CKD-related systemic inflammation impairs hypothalamic responses to appetite-regulating hormones, namely, leptin, ghrelin, and melanocortin, persistently activating the anorexic neural pathway [72,73]. Consequently, CKD patients develop anorexia that restricts the intake of amino acids and may reduce IGF-1 concentration, resulting in impaired protein synthesis and inhibition of muscle growth [5].

2.2.2. Oxidative stress

Oxidative stress occurs due to an imbalance between oxidation and anti-oxidation, which generally triggers the production of excessive reactive oxygen species (ROS) and reactive nitrogen species (RNS) [74,75]. Excessive ROS and RNS can lead to the oxidation of biomolecules, such as lipids, proteins, and DNA, causing certain harm to cells [27,76–78]. Mitochondria are an important source of ROS produced via normal metabolism in the body. Mitochondrial dysfunction and a reduced level of antioxidant enzymes increase ROS production. The activation of NADPH oxidase was also found to be involved in the ROS production pathways. Uremic toxins such as indoxyl sulphate contributes to renal fibrosis via OAT/NADPH Oxidase/ROS Pathway [79]. NADPH Oxidase Hyperactivity contributes to tissue injury through ROS-Mediated MAPK Signaling Pathway [80,81]. Increased ROS production and mitochondrial dysfunction are common in CKD, which usually trigger oxidative stress. Oxidative stress induces muscle atrophy in patients with advanced CKD [82]. Oxidative stress inhibits the AKT/mTOR pathway and its downstream targets, subsequently suppressing protein synthesis, which results in muscle atrophy in CKD patients [83,84]. In addition, local accumulation of ROS activates the autophagy pathway and induces muscle atrophy. As CKD progresses, excessive oxidative stress occurs with increased ROS production, thereby promoting the synthesis and release of pro-inflammatory cytokines [5], which may be associated with inflammation-mediated muscular atrophy, further exacerbating the extent of atrophy.

2.2.3. Mitochondrial dysfunction

Mitochondrial dysfunction activated proteolytic system may play an important role in the entire process of muscle atrophy [85]. Regulating mitochondrial biogenesis can improve resistance to muscle atrophy [18,86]. Mitochondrial dysfunction releases apoptosis-inducing factor (AIF) and cytochrome *c* into the cytosol, which leads to the activation of caspase-3. Cleaved caspase-3 promotes actin/myosin decomposition and induces myonuclear cell apoptosis [87]. Then Mitochondrial fission disrupts intracellular energy homeostasis, reduces ATP production and activates AMPK [88]. Activation of AMPK upregulates the expression of autophagy-specific gene proteins, which leads to the initiation of autophagy and ultimately to skeletal muscle atrophy [89]. The above research results indicate that mitochondrial dysfunction and oxidative stress can lead to CKD induced muscle atrophy in various ways.

2.3. Factors related to CKD-induced muscle atrophy

CKD-related complications, such as metabolic acidosis, vitamin D deficiency, anorexia, excess angiotensin II, as well as other related factors that play a role in CKD muscle atrophy, such as insulin resistance, hormones, hemodialysis, uremic toxins, intestinal flora imbalance, and miRNA (Fig. 2).

2.3.1. Metabolic acidosis

Metabolic acidosis is a complication of CKD, and it is a disorder of

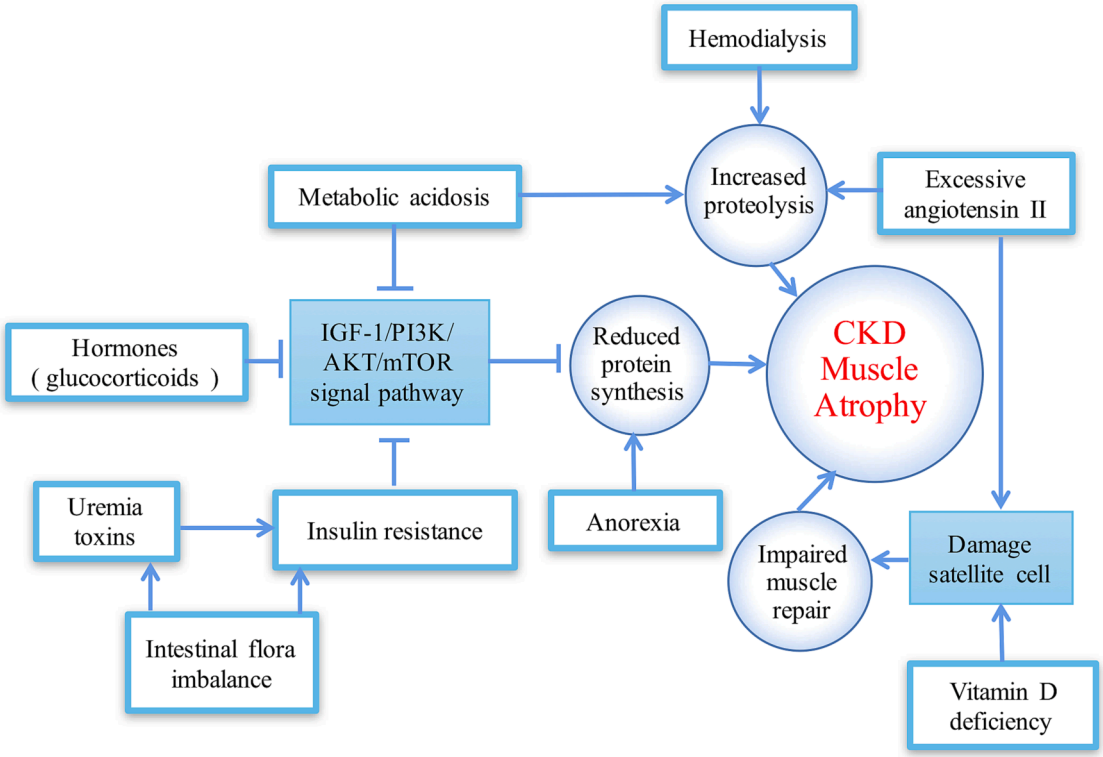

Fig. 2. Factors related to CKD-induced muscle atrophy.

acid-base balance that occurs when the bicarbonate concentration in plasma or venous blood is below 22 mM [90]. The occurrence of metabolic acidosis in CKD rats leads to a decrease in muscle protein storage, mainly because protein degradation is stimulated and protein synthesis is inhibited [55]. Studies have also shown that metabolic acidosis accelerates muscle atrophy by stimulating the UPS and inhibiting the IGF-1/PI3K/AKT signaling pathways [5,91]. Metabolic acidosis also induces muscle atrophy by stimulating the secretion of glucocorticoids from adrenal glands [5]. In conclusion, metabolic acidosis can trigger CKD-induced muscle atrophy in various ways, which provides important clues for the prevention and treatment of CKD-induced muscle atrophy in the future.

2.3.2. Vitamin D deficiency

CKD patients usually have vitamin D deficiency, which is now recognized as a complication of CKD [83]. Vitamin D regulates not only calcium homeostasis and bone metabolism but also skeletal muscle metabolism [92]. Vitamin D is involved in muscle cell differentiation and proliferation by binding to the vitamin D receptor, which is accompanied by interactions with muscle protein metabolism [93]. Vitamin D also affects the proliferation and differentiation of satellite cells [94,95]. However, a previous study has suggested that vitamin D deficiency is not the main factor leading to muscle mass loss in CKD patients [96]. Vitamin D deficiency may cause muscle atrophy by inhibiting myogenesis in CKD, but the specific mechanism remains to be further studied.

2.3.3. Anorexia

CKD patients often have a loss of appetite, and some patients even develop anorexia. The main causes of anorexia include disturbance in appetite-regulating hormones (such as leptin and ghrelin), decreased ability to discern taste, uremia-related gastrointestinal symptoms, depression, hydrodynamic instability due to the use of antihypertensive drugs or hemodialysis, and satiety during peritoneal dialysis [38]. In addition, reduced daily food intake results in malnutrition that reduces plasma albumin levels, thereby impairing muscle protein anabolism [97]. As mentioned above, inflammatory anorexia in CKD patients restricts the intake of amino acids and may reduce IGF-1 concentration, leading to impaired protein synthesis and thus, to some extent, accelerating muscle atrophy [5]. Therefore, muscle atrophy due to CKD is closely related to dietary nutrition.

2.3.4. Excessive angiotensin II

An elevated angiotensin II level is considered to be a complication of CKD. Angiotensin II is an endogenous peptide in skeletal muscle that promotes skeletal muscle atrophy. Angiotensin II also depolarizes membrane mitochondria, thereby increasing mitochondrial ROS production and causing oxidative stress in skeletal muscle [98]. Increased angiotensin level reduces the satellite cell pool and muscle regeneration ability [5].In addition, CKD activates the renin-angiotensin-system, which activates caspase 3 in skeletal muscle and induces actin cleavage [38,99], thereby triggering muscle atrophy. Angiotensin II also increases muscle proteolysis via activating UPS and TGF-β pathway and reduces circulating IGF-1 levels, which is a major mechanism of muscle mass loss [38,100]. In conclusion, CKD causes excessive angiotensin II in the body leading to skeletal muscle atrophy.

2.3.5. Insulin resistance

Insulin resistance is a pathological condition in which target tissues fail to respond normally to insulin [101]. CKD has been confirmed to cause insulin resistance [102]. As a result of renal insufficiency, insulin resistance is common in patients with renal diseases, and its prevalence increases, especially in advanced stages of CKD. Insulin resistance worsens as renal function declines. Insulin signaling deficiency is associated with the loss of muscle mass, and insulin resistance in CKD stimulates muscle atrophy. Under normal conditions, insulin acts as a powerful synthetic signal that binds to insulin receptors to induce tyrosine autophosphorylation of IRS-1, thereby stimulating muscle protein synthesis [103]. However, inhibition of insulin signaling downregulates the PI3K/AKT pathway and stimulates the autophagy

system via lysosomes, ultimately reducing protein synthesis and increasing muscle protein degradation [104]. In addition, tyrosine-protein phosphatase nonreceptor type substrate 1 also contributes to CKD-induced muscle protein depletion by reducing tyrosine phosphorylation of insulin receptors and IRS-1 as well as aggravating insulin resistance and stimulating muscle atrophy [55]. Toll-like receptors (TLRs) modulate local innate immune responses. TLR13 is involved in CKD-induced insulin resistance in muscle tissue, and TLR13 increases serine phosphorylation and decreases AKT phosphorylation through the recruitment of interferon regulatory factor 3 (IRF3). The decreased AKT phosphorylation mediates CKD-induced muscle atrophy by stimulating protein degradation through the UPS. Targeting TLR13 improves insulin sensitivity and prevents muscle atrophy under catabolic conditions in which insulin signaling is damaged [105]. Therefore, insulin resistance is an important factor affecting muscle atrophy in CKD patients.

2.3.6. Hormones

Several hormones can also trigger sarcopenia. Among these hormones, glucocorticoids induce proteolysis by increasing the expression of myostatin and reduce of protein synthesis by inhibiting the mTOR pathway [106]. Glucocorticoids inhibit the production of IGF-1 and accelerate the degradation of IRS-1, which reduces the activity of PI3K, thereby inhibiting protein synthesis and ultimately promoting muscle atrophy. In addition, Growth hormone (GH) is also an important factor for muscle gain. Studies have shown that the abnormal GH/IGF-1 physiological axis is considered to be the underlying cause of increased protein catabolism and CKD-induced muscle atrophy [38,107]. Skeletal muscle is the target tissue of androgens, which exert beneficial effects on muscle mass, strength, and regeneration [108]. Testosterone is an important male hormone for muscle hypertrophy. Testosterone promotes muscle anabolism, stimulates protein synthesis, inhibits muscle degradation, and improves muscle reutilization efficiency of amino acids [5,109]. Over 60 % of patients with advanced CKD have evidence of low serum testosterone levels, and low plasma testosterone levels may contribute to muscle loss due to altering IGF-1 signaling and increasing myostatin concentrations [109]. Additionally, female patients with CKD exhibit estrogen deficiency in the early stage of the disease, and reduced estrogen levels are associated with reduced muscle strength and function [5]. In a mouse model of CKD, parathyroid hormone/parathyroid hormone receptor signals in adipose tissue trigger inappropriate thermogenesis and hypermetabolism, leading to muscle atrophy [110]. Therefore, human hormone levels also contribute to muscle atrophy in CKD patients.

2.3.7. Hemodialysis

Muscle atrophy occurs in CKD patients, and hemodialysis increases the loss of muscle mass in CKD patients [5]. The UPS and ALP are simultaneously activated in chronic hemodialysis patients, and some E3 ubiquitin ligases act as the key markers of human muscle atrophy [111]. Upregulated key autophagy-related proteins in hemodialysis patients are considered to be potent and durable markers of muscle atrophy [111]. On the hand, dietary proteins/amino acids and physical activity are the two major anabolic stimuli that increase muscle protein synthesis in CKD patients [112]. However, the patients are exposed to amino acid deprivation during the dialysis sessions [113]. Furthermore, undergoing hemodialysis have reduced levels of physical activity or even no exercise at all, which is also a determinant of muscle protein loss and muscle atrophy [114,115]. Hemodialysis also affects the muscle mass by the maintenance of inflammation due to the biocompatibility of the materials [116]. Hemodialysis is an influencing factor of CKD-induced muscle atrophy, and the relationship and specific mechanisms between them need to be further studied.

2.3.8. Uremic toxins

Accumulation of uremic toxins has been widely considered to be the main trigger of CKD-induced muscle atrophy. For example, methylglyoxal is a highly reactive dicarbonyl uremic toxin that accumulates in the circulatory system of CKD patients, and it has been implicated in the pathology of uremic sarcopenia. Methylglyoxal induces muscle atrophy, inflammation, fibrosis, and oxidative stress in myoblasts. Methylglyoxal-induced intramuscular metabolic changes trigger ATP shortage in myoblasts, which is a pathogenic factor in the development of CKD-related sarcopenia [117]. Indoxyl sulfate (IS) acts as a representative protein-bound uremic toxin that significantly accumulates in the plasma and skeletal muscle of CKD patients. IS is an indole-containing compound that may cause myotube atrophy in C2C12 cells by increasing oxidative stress and MAPK activation [118]. IS stimulates ROS-mediated up-regulation of myostatin and MAFbx, and it also accelerates mitochondrial dysfunction by inducing oxidative stress in skeletal muscle cells [119]. Impaired mitochondrial function then leads to reduced muscle mitochondrial mass, reduced skeletal muscle energy production, and oxidative damage to mitochondrial components, thereby inducing ROS generation, resulting in the formation of a vicious circle, which leads to the exacerbation of skeletal muscular dystrophy [82]. In addition, uremic toxins accumulate in the body of CKD patients, which induces insulin resistance and mitochondrial energy metabolism impairment, further triggering skeletal muscle atrophy [102]. These close relationships provide new insights into the mechanism by which uremic toxins induce muscle atrophy in CKD.

2.3.9. Intestinal flora imbalance

The human gut microbiota is composed of more than 100 trillion microorganisms that coexist with the host and has various regulatory effects on health maintenance, including nutrient metabolism, immune regulation, energy homeostasis, and external detoxification as well as biosynthesis of vitamins, hormones and neurotransmitters [120]. The causal relationship between gut dysbiosis and muscle atrophy is mediated by reduced bioavailability of dietary amino acids, chronic systemic inflammation, insulin resistance, mitochondrial dysfunction, and modulation of host gene expression [120,121]. Intestinal flora imbalance can lead to the failure of intestinal barrier function, resulting in increased intestinal permeability and immune-mediated inflammatory response [122]. Studies have shown that uremic dysbiosis induces micro-inflammation by increasing intestinal tract permeability, changing the serum concentrations of uremic solutes and increasing additional levels of bacterial fermentation products. These changes induce insulin resistance and mitochondrial dysfunction, ultimately causing sarcopenia [51,123]. There is an interaction between gut microbiota and CKD. CKD may affect the structure of the gut microbiota and lead to gut dysbiosis, while gut dysbiosis may increase the level of uremic toxins in CKD patients, thereby leading to the progression of CKD [124]. In addition, the retention of uremic toxins caused by gut dysbiosis triggers oxidative stress and endothelial dysfunction, which may also promote muscle atrophy [125].

2.3.10. MiRNAs

MiRNAs are a group of short non-coding RNAs that are widely involved in the regulation of skeletal muscle homeostasis, growth, regeneration, metabolism and atrophy [51,126]. In recent years, miRNAs have been confirmed to participate in the regulation of CKD-induced muscle atrophy by increasing protein degradation or impairing myogenesis [126]. In the context of CKD, skeletal muscle atrophy is associated with a decrease in miR-26a, and miR-26a overexpression relieves CKD-induced muscle atrophy by inhibiting the FOXO1 transcription factor [127,128]. A decreased level of miR-29 leads to the upregulation of YY1 transcription factor activity, thereby inhibiting myogenesis, while overexpression of miR-29 alleviates CKD-induced muscle atrophy by downregulating YY1 and TGF-β pathway proteins [55,129,130]. miRNA-23a inhibits the expression levels of MuRF1 and MAFbx, and miRNA-27a inhibits FoxO and reduces myostatin levels to stimulate muscle cell proliferation. Both miRNA-23a and miRNA271 are significantly reduced in CKD patients, which enhances protein

degradation [55,126,131]. In addition, overexpression of miRNA-23a and miRNA-27a reduces myostatin expression, inhibits downstream SMAD-2/3 signaling, and reduces the activation of caspase 3 and caspase 7 as well as increases the expression of muscle regeneration markers [131]. miRNA-206 and miRNA-486 promote myoblast differentiation by downregulating Pax7 [55]. Overall, these experimental data increase our understanding of the role of miRNAs in CKD-induced muscle atrophy, suggesting that miRNAs may serve as potential therapeutic targets for the treatment of CKD-induced muscle atrophy.

3. Treatment FOR CKD-INDUCED muscle atrophy

To treat CKD-induced muscle atrophy, CKD must be treated first by interventions targeting the related mechanisms of CKD-induced muscle atrophy, and appropriate treatment strategies and drugs should then be used to delay or alleviate CKD-induced muscle atrophy. The treatment for CKD is mainly to actively control the level of proteinuria, control blood pressure, delay the progression of CKD, and help improve the conditions and quality of life of patients. As the disease progresses, patients with CKD may develop symptoms, such as hypertension, anemia, oliguria, and cardiac insufficiency, indicating that patients should be systematically checked and given medication for the corresponding complications. In addition, for non-drug treatments, patients with CKD should have a low-salt, low-fat, and high-quality protein diet, which encourages patients to consume limited amounts of sodium, protein, cholesterol, and potassium but increase consumption of fiber and vitamin-rich foods [125]. Proper exercise is also recommended to strengthen the patient's body. When CKD progresses to the end stage, renal replacement therapies, such as hemodialysis, peritoneal dialysis, and kidney transplantation, can be performed. Early prognosis and treatment should be recommended for CKD prevention and treatment. At present, there is no cure for CKD in most patients. Therefore, the goal of CKD treatment should be to control complications and slow disease progression, avoiding its progression to the uremic stage (Fig. 3).

3.1. Rectify metabolic acidosis

Chronic metabolic acidosis can cause muscle catabolism in patients with CKD. Over the past 20 years, the treatment of metabolic acidosis is recommended to counteract the detrimental effects of metabolic acidosis on bone and muscle in clinical practice. Metabolic acidosis is usually treated with oral alkali supplements or methods that reduce dietary acid intake in patients with CKD. Oral alkali supplementation or reduced dietary acid intake may slow the decline in renal function, improve serum bicarbonate levels, and reduce the risk of progression to end-stage kidney disease in patients with CKD and metabolic acidosis [132]. In addition, alkaline treatment with sodium bicarbonate increases insulin sensitivity [90], which may contribute to slowing muscle atrophy. Veverimer, a new drug for treating metabolic acidosis, is a novel non-absorbable polymer, which selectively binds to and eliminates hydrochloric acid in the gastrointestinal tract rather than using a base, such as in bicarbonate or citrate formulations, and then treats metabolic acidosis through binding to and eliminating protons in the gut [133]. A recent study has shown that veverimer effectively treats metabolic acidosis in female CKD patients by reducing protein degradation, which is beneficial to relieve muscle atrophy [134]. However, there are still many problems in the current treatment of metabolic acidosis, such as lack of approved drugs, poor dietary compliance, and adverse side effects due to currently available oral alkali supplements [135]. Therefore, further investigations are required.

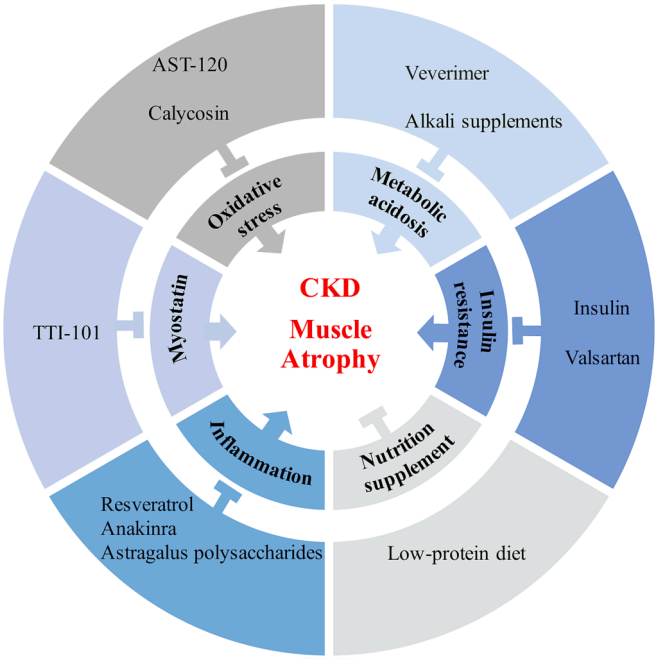

Fig. 3. Drugs for chronic kidney disease-induced muscle atrophy.

3.2. Reduce insulin resistance

For patients with CKD, previous guidelines have recommended the use of repaglinide or dipeptidyl peptidase 4 inhibitor combined with additional insulin to treat insulin resistance [136,137]. As the use of insulin can cause body mass gain and hypoglycemia, the additional use of GLP1-RA or SGLT2i is also proposed to reduce the dose of insulin [136]. Chronic renal failure may also induce insulin resistance in skeletal muscle, but valsartan can partially alleviate this phenomenon [138]. As an angiotensin receptor blocker, valsartan acts on angiotensin II receptor 1 to overall enhance the activity of IRS-1 and its downstream insulin signaling pathway through the common pathway as well as to alter the abnormal phosphorylation of important molecules in the IRS/PI3K/AKT pathway, thereby alleviating insulin resistance [138]. At present, there are few studies on the treatment of CKD-induced muscle atrophy by targeting insulin resistance, which may be a new research direction for the treatment of CKD-induced muscle atrophy.

3.3. Nutritional supplement of low-protein diet

A low-protein diet has long been an important measure to delay the progression of CKD. The rationale of a low-protein diet is based on long-term control of dietary protein intake, which effectively alleviates uremia symptoms and metabolic acidosis as well as the further progression of renal function damage. However, a low-protein diet often leads to malnutrition [97], further leading to other adverse events including muscle atrophy. For those patients receiving hemodialysis have high-quality protein diet protein requirements, as they experience protein losses during dialysis [139]. For CKD patients who have adopted a low-protein diet for a long time, malnutrition can be prevented by supplementing essential amino acids (α-keto acids) and some trace elements. In addition, ketoacid supplementation improves muscle atrophy, oxidative damage, and mitochondrial dysfunction in a rat model of CKD + low-protein diet [17]. Another study has also indicated that ketone analog supplementation also helps prevent loss of skeletal muscle mass in patients with CKD [140]. The use of nutritional supplements also directly provides energy for muscle activity and helps maintain muscle mass, especially for the patients on hemodialysis [139,141]. Therefore, scientific and reasonable low-protein diets are extremely important for patients with CKD.

3.4. Other drugs

In a CKD model, inhibition of myostatin prevents muscle atrophy,

increases the function of satellite cells, improves IGF-1 signaling, and inhibits protein degradation. To date, however, there is no clinically effective myostatin inhibitor. A recent study has identified a STAT3/CCAAT enhancer-binding protein delta and myostatin signaling pathway, which activates muscle protein degradation in CKD mice, and a small molecule inhibitor of STAT3 (TTI-101) has been found to block this pathway and improve muscle mass [142]. Resveratrol is a natural polyphenol with strong biological activity, and it exerts beneficial effects to suppress the upregulation of MuRF1 and relieves muscle atrophy by attenuating the degradation of inhibitor of nuclear factor kB alpha and inhibiting the activation of nuclear factor-kB [143]. In addition, resveratrol improves CKD-induced muscle atrophy through the SIRT1/FoxO1 axis, indicating that resveratrol combined with SIRT1 agonists also has potential for the treatment of CKD and CKD-induced muscle atrophy [144]. Calycosin is the main component of astragalus, and it has anti-inflammatory, antioxidant, and anti-autophagic effects. Calycosin inhibits oxidative stress and autophagy in CKD-induced skeletal muscle atrophy and TNF-α-induced C2C12 myotube atrophy, and it also reduces MuRF1 and MAFbx levels to alleviate muscle atrophy [145]. Uremia toxin IS induces mitochondrial dysfunction by increasing oxidative stress to accelerate skeletal muscle atrophy. Based on this molecular mechanism, AST-120, which inhibits IS accumulation, can be used for IS-targeted intervention. Mitochondrial-targeted interventions with L-carnitine or teneligliptin (a dipeptidyl peptidase 4 inhibitor) can be used to improve muscle atrophy and athletic ability [146]. Additionally, androgenic steroids, such as nandrolone decanoate (a synthetic derivative of testosterone), have also been reported to increase muscle mass in healthy adults and patients with CKD [5]. Astragalus polysaccharides may reduce the expression of MAFbx by inhibiting the nuclear factor-κB pathway to ultimately delay or alleviate muscle cell atrophy associated with chronic renal failure and cachexia [147]. Systemic inflammation is associated with excessive skeletal muscle atrophy in patients with CKD as well as in animal models of CKD, suggesting that anti-cytokine therapy may be a potential strategy for the treatment of CKD-related cachexia. Anakinra acts as an IL-1 receptor antagonist to reduce the levels of IL-6, TNF-α, and IL-1β in serum and muscle of wild type (WT)/CKD mice, indicating that IL-1 receptor antagonists may represent a new targeted therapy for CKD-induced muscle atrophy [148]. Skeletal muscle atrophy is an important clinical feature of CKD. The first-line therapy for CKD-induced muscle atrophy is to reduce CKD-related complications, such as metabolic acidosis, elevated angiotensin II, poor control of diabetes, and accumulation of unexcreted metabolic waste products [55]. However, the current treatment for CKD-induced muscle atrophy is still in the early stage of development, thus requiring additional research.

3.5. Physical exercise and its auxiliary means

Low physical activity is most common in patients with CKD, which leads to low exercise capacity. Appropriate physical exercise is still the primary method for increasing skeletal muscle mass and treating sarcopenia [40,149]. Exercise enhances IGF-1/AKT/mTOR signaling in CKD mice and patients with CKD, thereby reducing muscle loss [126]. Weekly resistance training may be an effective rehabilitation strategy for patients with stage 3 CKD [150]. Resistance training has been found to reduce renal fibrosis as well as inflammation in CKD rats [151]. Moderate-intensity exercise training also reduces the plasma level of myostatin in patients with CKD [152]. For patients with non-dialysis dependent CKD, aerobic exercise (treadmill, rowing, or cycling) with resistance exercise (leg stretches and leg presses) three times per week for 12 weeks is better than aerobic exercise alone to increase muscle mass and strength [153]. However, for some CKD patients who are frail and unable to exercise voluntarily, neuromuscular electrical stimulation and acupuncture combined with low-frequency electrical stimulation is an alternative to physical exercise [5]. In addition, respiratory muscle training, yoga during dialysis, and blood flow restriction training can also improve low physical activity in patients with CKD [5,154]. In conclusion, low physical activity in patients with CKD impacts their health and quality of life. Comprehensive intervention strategies are preferred, and the unique pathophysiology of CKD and the individual conditions of patients with CKD should also be considered.

4. Perspectives

CKD is a common progressive disease worldwide, and muscle atrophy is one of its complications, which seriously impacts the body function and quality of life of CKD patients. This review briefly introduces the mechanism of muscle atrophy and the harm of CKD-induced muscle atrophy, and it summarizes the molecular mechanism of CKD-induced muscle atrophy and some corresponding treatment methods. Muscle atrophy and sarcopenia in CKD patients have complex pathways that involve multiple mechanisms and relevant factors. In addition to some known factors, such as insulin resistance, autophagy, myostatin, uremic toxins, inflammation, and oxidative stress, the mechanism of CKD-induced muscle atrophy is also greatly affected by CKD-related complications, such as metabolic acidosis, vitamin D deficiency, and elevated angiotensin II. Based on these findings, potential therapeutic drugs against CKD-induced muscle atrophy can be developed. In addition, miRNAs may also serve as potential therapeutic targets for CKD-induced muscle atrophy. Furthermore, recent advances show that muscle produces myokines in response to exercise, which allow for crosstalk between the muscle and other organs [155]. Due to muscle–kidney crosstalk, treatment of muscle atrophy is accompanied by benefits in kidney. For example, the agents promoting the secretion of irisin, a myokine, could be an underlying therapeutic strategy in alleviating the progression of CKD [156]. Learning more about the mechanisms of CKD-induced muscle atrophy is crucial for developing new clinical therapeutic strategies. The treatments and strategies mentioned in this review are expected to be effective treatments for CKD-induced muscle atrophy and even for CKD. As increasing evidence uncovers the pathways leading to muscle atrophy, new interventions for muscle atrophy are expected to be developed, thus providing a better quality of life and higher survival rates for patients with CKD.

CRediT authorship contribution statement

Kexin Wang: Methodology, Resources, Data curation, Writing – original draft, Writing – review & editing, Visualization. **Qingyuan Liu:** Methodology, Resources, Data curation, Writing – original draft, Writing – review & editing. **Mingyu Tang:** Methodology, Resources, Data curation, Writing – original draft. **Guangdong Qi:** Methodology, Data curation. **Chong Qiu:** Methodology, Data curation. **Yan Huang:** Methodology, Data curation. **Weiran Yu:** Data curation. **Wei Wang:** Data curation. **Hualin Sun:** Writing – review & editing, Funding acquisition. **Xuejun Ni:** Conceptualization, Writing – review & editing, Project administration. **Yuntian Shen:** Conceptualization, Writing – original draft, Writing – review & editing, Project administration, Funding acquisition. **Xingxing Fang:** Conceptualization, Writing – review & editing, Project administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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